Madhab Chandra Das

Curriculum Vitae for Digital Forensics and Cybersecurity

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Professional Summary

Digital and cyber forensics PhD student with a master's degree in Cybersecurity and hands-on training in Windows artifact analysis, forensic imaging, deleted file recovery, packet analysis, and e-discovery. Uses Python, Autopsy, FTK Imager, Wireshark, TShark, ALEAPP, and Volatility in coursework and research. Builds structured datasets and evaluates machine learning and LLM outputs against evidence. Brings IT support experience and banking experience in due diligence, customer problem solving, staff training, and leadership of teams of 11–15 members.

Technical Skills

Forensic tools: Autopsy, FTK Imager, HxD, John the Ripper, Wireshark, TShark, ALEAPP, Volatility 2 and 3.

Programming and systems: Python, PowerShell, SQL, Linux, VirtualBox, Kali Linux, Parrot OS.

AI and analysis: Random Forest, XGBoost, LightGBM, CatBoost, Logistic Regression, SHAP, permutation importance, LLM APIs.

E-discovery: PST Analyzer, MBOX Viewer, GoldFynch. Productivity: Microsoft Excel, PowerPoint and Word.

Practical Forensic Training

Windows and Operating System Forensics

Created forensic images, recovered deleted files, examined files and metadata, and performed hex analysis and password recovery. Analyzed Registry, event logs, application and user activity, USB artifacts, email headers, and browser artifacts, including LevelDB, IndexedDB, cache, cookies, history, and HAR records.

Network Forensics

Used Wireshark and TShark to examine TCP connections, handshakes, visible payloads, login-related information, and suspicious traffic. Identified IP addresses associated with potentially malicious activity.

Mobile and Memory Forensics

Used ALEAPP for Android artifact analysis and Linux scripting for memory analysis. Research also used Volatility 2 and 3, with Volatility 2.6 supplying the final process-feature dataset.

E Discovery

Completed case-based exercises involving electronic records and preservation. Studied TAR, METS, the Federal Rules of Civil Procedure and Criminal Procedure. Used PST Analyzer, MBOX Viewer, and GoldFynch.

Education

PhD in Digital and Cyber Forensics | Sam Houston State University

August 2025–Present | Expected July 2029 | Current CGPA: 4.00

Master's in Cybersecurity | Washington University of Science and Technology

October 2023–June 2025 | CGPA: 3.92/4.00

MBA | Jahangirnagar University | August 2020–July 2022

BSc in Electrical and Electronic Engineering | UITS | August 2008–January 2013

Academic Experience

Graduate Research Assistant and Teaching Assistant

Sam Houston State University, Department of Computer Science | August 2025–Present

- Conduct forensic research using Python, artifact examination, structured datasets, and evaluation of AI and machine learning outputs.
- Conduct lab classes, proctor examinations, grade exam responses, and provide tutoring outside scheduled class hours.

Courses supported

DFSC 3316 — Cryptography and Network Security | Spring and Fall 2026
COSC 1437 — Programming Fundamentals II | Fall 2025
DFSC 2316 — DF and IA Fundamentals II | Fall 2026

Research Experience

Android Forensic Reporting

Examined public Android 13 data and a physical LG device using Python and ALEAPP. Generated reports through three LLM APIs and evaluated 612 claims across six reports against verified findings.

Network Traffic Analysis

Prepared two encrypted traffic datasets and evaluated four ML models. Analyzed 41,238 external false positives (4.35% FPR) using SHAP and permutation importance.

Windows Memory Forensics

Built 1,899 process records with 18 features from 45 memory dumps using Volatility 2.6. Evaluated three ensembles with dump-grouped validation against rule-derived labels.

Quantum Security and Forensics

Generated two physics-inspired synthetic QKD datasets with 5 million records each. Implemented classifiers and examined reliability under changed simulation conditions.

Mobile Screenshot Forensics

Contributed literature review and manuscript writing to ScreenForensics.

Additional Professional Experience

The City Bank Ltd., Bangladesh | March 2017–July 2022

Associate Manager | December 2021–July 2022

Assistant Manager | January 2019–November 2021

Senior Officer and Team Leader | May 2018–January 2019

SPMO and GL, contractual | March 2017–May 2018

Experience in banking due diligence, customer relationships, staff training, and leadership of 11–15 team members.

Mutual Trust Bank Ltd. | Senior Sales Officer, contractual | September 2016–February 2017

BRAC Bank Ltd. | Trainee Officer, contractual | June 2013–June 2016

BRAC Bank Ltd. | Technology Intern | June–September 2011

Internship support included Windows reinstallation, application troubleshooting, hardware, printers, and document archiving.

Relevant Graduate Coursework

Completed with A grades: Operating System Forensics, Network Forensic Analysis, Live System & Memory Forensics, E-Discovery, Mobile Device Forensics, DF Laboratory Management, Cryptography & Steganography, and Scientific Communications.

In progress, Fall 2026: File System Forensics and Computational Forensics.

Training and Professional Development

Cybersecurity and Cloud

Cybersecurity and Risk Management Specialist — PIIT | 2024

Getting Started with AWS Cloud Essentials — AWS through W3Schools | 2024

Learning Cryptography and Network Security — LinkedIn Learning | 2026

Wireshark: Functionality — LinkedIn Learning | 2026

Artificial Intelligence and Quantum Computing

Artificial Intelligence Training — SHSU | 2026

Introduction to Quantum Computing — LinkedIn Learning | 2026

Peer Review Training

An Introduction to Peer Review — Clarivate | 2025

Certified Peer Reviewer Course — Elsevier Researcher Academy | 2025

Practical Participation

Capture the Flag Security Event — Cisco–SHSU | February 2026

Honors and Leadership

- COSET PhD Digital & Cyber Forensics Scholarship, SHSU, 2025–2026. Renewed for 2026–2027.
- WUST Presidential Honor: Summer 2024, Fall 2024, and Spring 2025.
- President, Bangladesh Student Association, SHSU | August 2025–Present. Organize events and support incoming students.

Selected Publications

Accepted

Madhab Chandra Das, Sajid Ahmed, Tasnia Tabassum, and Qingzhong Liu. “Evidence-Grounded LLM-Assisted Android Mobile Forensic Reporting.” IEEE TPS 2026. Accepted.

Madhab Chandra Das and Qingzhong Liu. “Explaining Cross-Dataset False Positives in Encrypted Traffic Classification for Forensic Interpretation.” IEEE UEMCON 2026. Accepted.

Madhab Chandra Das and Amar Rasheed. “Rule-Derived Process Classification in Windows Memory Forensics: A Reliability-Oriented Evaluation.” ISDFS 2027. Accepted.

Under Review

Madhab Chandra Das and Qingzhong Liu. “Investigator-Centered Real-Time IoT IDS Using ML and XAI: A Systematic Survey and Framework.” ACM Computing Surveys. Under review.